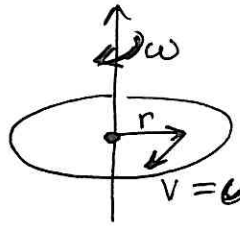
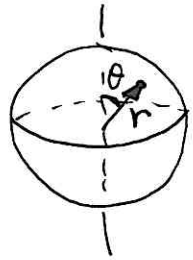
P5.6

a)




$v = \omega r \Rightarrow K = \sigma v = \sigma \omega r$

b)




$\vec{J} = \rho \vec{v}$ $\quad \vec{\omega} = \omega \hat{z} = \omega(\cos\theta\, \hat{r} - \sin\theta\, \hat{\theta})$

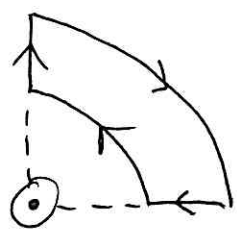
$\vec{v} = \vec{\omega} \times \vec{r}$

$\rho = \dfrac{Q}{\frac{4\pi}{3}R^3}$ $\quad \Rightarrow \vec{v} = -\omega \sin\theta\, r\, \underbrace{\hat{\theta} \times \hat{r}}_{-\hat{\varphi}} = \omega r \sin\theta\, \hat{\varphi}$

$$\vec{J} = \frac{Q}{\frac{4\pi}{3}R^3}\, \omega r \sin\theta\, \hat{\varphi}$$

P5.9

a)



$$\vec{B}(\vec{r}) = \frac{\mu_0 I}{4\pi} \int \frac{d\vec{l} \times (\hat{r} - \hat{r}')}{|\vec{r} - \vec{r}'|^2}$$

$d\vec{l} \times \hat{r} \Rightarrow \hat{z}$ = normal vector out of page

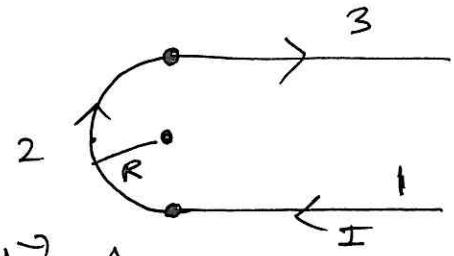
$$\vec{B}_a = \frac{\mu_0 I}{4\pi}\, \hat{z} \int_0^{\pi/2} \frac{a\, d\varphi}{a^2} = \frac{\mu_0 I}{8a}\, \hat{z}$$

$$\vec{B}_b = \frac{\mu_0 I}{4\pi}\, \hat{z} \int_{\pi/2}^{0} \frac{b\, d\varphi}{b^2} = -\frac{\mu_0 I}{8b}\, \hat{z}$$

Straight segments do not contribute.

$$\vec{B} = \frac{\mu_0 I}{8}\left(\frac{1}{a} - \frac{1}{b}\right) \hat{z}$$

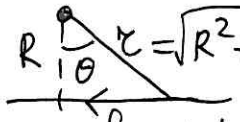
f)




$\vec{B}$ is into the page.

$d\vec{l} \times \hat{r}$ is into the page $= (-\hat{z}) \cdot \sin\left(\frac{\pi}{2}+\theta\right) = (-\hat{z}) \cos\theta$

segment 1: $\vec{B}_1 = \frac{\mu_0 I}{4\pi}(-\hat{z}) \int_{-\pi/2}^{0} \cos\theta \frac{R\,d\theta}{\cos^2\theta} \frac{1}{(R/\cos\theta)^2} =$



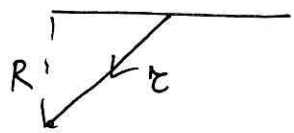

$r = \sqrt{R^2+l^2} = \frac{R}{\cos\theta}$

$l = R\tan\theta \Rightarrow dl = \frac{R\,d\theta}{\cos^2\theta}$

$= \frac{\mu_0 I}{4\pi R} \sin\theta \Big|_{-\pi/2}^{0} (-\hat{z})$

$= \frac{\mu_0 I}{4\pi R}(-\hat{z}).$

segment 3:




$\vec{B}_3 = \vec{B}_1$

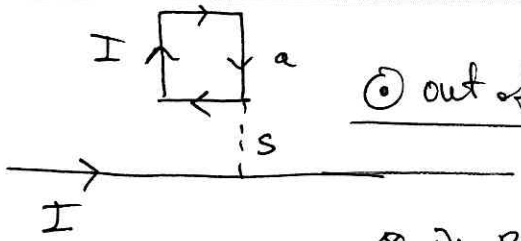
segment 2: $\vec{B}_2 = \frac{\mu_0 I}{4\pi}(-\hat{z}) \int_0^{\pi} \frac{R\,d\varphi}{R^2} = \frac{\mu_0 I}{4R}(-\hat{z})$

$\Rightarrow \vec{B} = \vec{B}_1 + \vec{B}_2 + \vec{B}_3 = (-\hat{z}) \left( \frac{2\mu_0 I}{4\pi R} + \frac{\mu_0 I}{4R} \right)$

$= (-\hat{z}) \frac{\mu_0 I}{4R}\left(1 + \frac{2}{\pi}\right).$

P5.10

a)

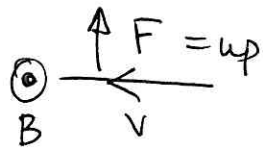



$\odot$ out of page $= \hat{z}$

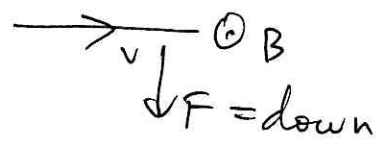
$\otimes$ in page

Mag. field produced by straight wire $B = \dfrac{\mu_0 I}{2\pi \rho}$

Lowerside: $\vec{v} \times \vec{B}$




Upper side: $\vec{v} \times \vec{B}$

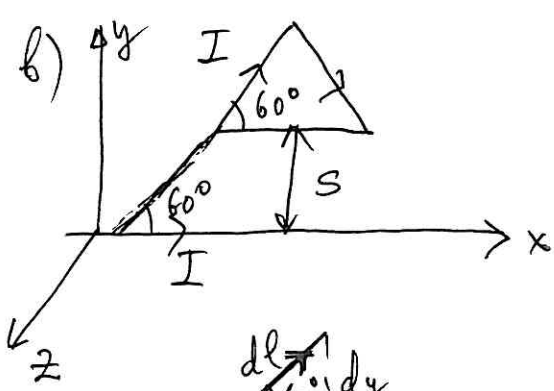



Force on the current loop:

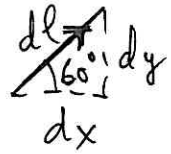
$$F = \underbrace{\left(\frac{\mu_0 I}{2\pi s}\right) I a}_{\text{lower}} - \underbrace{\left(\frac{\mu_0 I}{2\pi (s+a)}\right) I a}_{\text{upper}} =$$

$$= \frac{\mu_0 I^2 a}{2\pi}\left(\frac{1}{s} - \frac{1}{s+a}\right) = \frac{\mu_0 I^2 a^2}{2\pi s(s+a)}.$$

b)




$$\vec{F}_{\text{Bottom}} = \frac{\mu_0 I^2 a}{2\pi s}\,\hat{y}$$

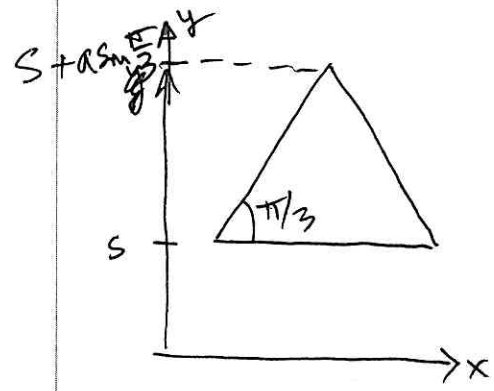



$$\vec{F}_{\text{left}} = I \int d\vec{l} \times \vec{B}$$

$$d\vec{l} = \frac{dl}{\cos\frac{\pi}{3}}\,\hat{x} + \frac{dl}{\sin\frac{\pi}{3}}\,\hat{y}$$

$$\vec{F}_{right} = I \int d\vec{l} \times \vec{B} \quad , \quad d\vec{l} = \frac{dl}{\cos\frac{\pi}{3}}\hat{x} - \frac{dl}{\sin\frac{\pi}{3}}\hat{y}$$

$$\Rightarrow \vec{F}_{left+right} = I \int \left(\frac{2\,dl}{\cos\frac{\pi}{3}}\hat{x}\right) \times \left(\frac{\mu_0 I}{2\pi y}\hat{z}\right)$$




$$\left.\begin{aligned} dx &= \frac{dl}{\cos\frac{\pi}{3}} \\ dy &= \frac{dl}{\sin\pi/3} \end{aligned}\right\} \quad \frac{dy}{dx} = \frac{\cos\frac{\pi}{3}}{\sin\frac{\pi}{3}} = \frac{1/2}{\sqrt{3}/2} = \frac{1}{\sqrt{3}}$$

$$dx = \sqrt{3}\,dy$$

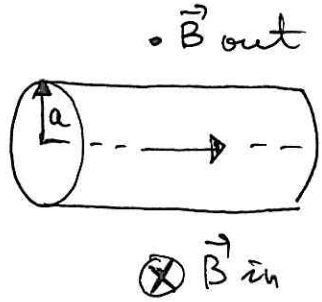
$$\vec{F}_{l+r} = 2I\left(\frac{\mu_0 I^2}{2\pi}\right)\underbrace{(\hat{x}\times\hat{z})}_{-\hat{y}} \int_s^{s+a\sin\frac{\pi}{3}} \frac{\sqrt{3}\,dy}{y} =$$

$$= -\frac{\mu_0 I^2}{\pi}\hat{y}\,\sqrt{3}\,\ln\left(\frac{s+\frac{\sqrt{3}}{2}a}{s}\right)$$

$$\vec{F}_{total} = \left\{\frac{\mu_0 I^2 a}{2\pi s} - \frac{\sqrt{3}\mu_0 I^2}{\pi}\ln\left(1+\frac{\sqrt{3}a}{2s}\right)\right\}\hat{y}.$$

P 5.13

a)




$\vec{J}$ = const on the outside surface of the wire

$\Rightarrow$ we have surface current $K = \frac{I}{2\pi a}$

By symmetry $\vec{B}$ circles around the wire.

$$\oint \vec{B}\cdot d\vec{l} = \mu_0 I_{enc} = \mu_0 \begin{cases} 0 & \text{for } r<a \\ I & \text{for } r>a \end{cases}$$

$$\oint \vec{B}\cdot d\vec{l} = B\cdot 2\pi r$$

$$\Rightarrow B = \begin{cases} 0 & \text{for } r<a \\ \frac{\mu_0 I}{2\pi r}\,\hat{\varphi} & \text{for } r>a \end{cases}$$

b) $J = ks$ ($s$ = distance from the wire axis)

total $I = \int J\,da = \int_0^a ks\cdot 2\pi s\,ds = 2\pi k\int_0^a s^2 ds = \frac{2\pi k}{3}a^3$

$$\Rightarrow k = \frac{3I}{2\pi a^3}.$$

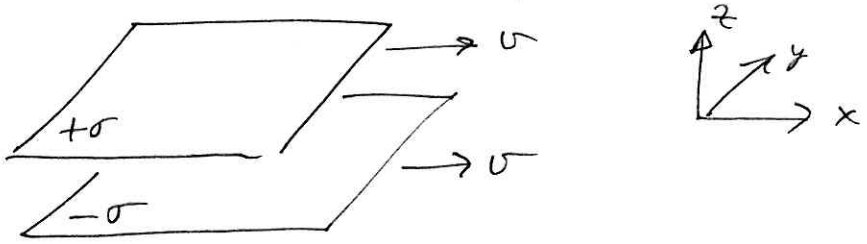
$$\Rightarrow B\cdot 2\pi s = \cancel{\mu_0 \int_0^s dr\, kr = \mu_0 k \frac{s^2}{2}}\ \mu_0 I_{enc}$$

$$I_{enclosed} = \int_0^s kr\cdot 2\pi r\,dr = k\,2\pi\frac{s^3}{3}.$$

$$\Rightarrow B(s) = \mu_0 k\frac{s^2}{3} = \frac{\mu_0 I}{2\pi a^3}s^2 \text{ for } s<a.$$

$$B(s) = \frac{\mu_0 I}{2\pi s} \text{ for } s>a \text{ of course.}$$

P5.16




(a) From Example 5.8: uniform current $\vec{K} = K\hat{x}$

produces $\vec{B} = \begin{cases} \frac{\mu_0}{2} K \hat{y} & \text{for } z<0 \\ -\frac{\mu_0}{2} K \hat{y} & \text{for } z>0 \end{cases}$

field of a top plate

Note that $\begin{cases} \text{top plate produces } K_t = \sigma v \\ \text{bottom plate} \quad K_b = -\sigma v \end{cases}$

Thus $\vec{B}_{\text{bottom}} = \begin{cases} +\frac{\mu_0 \sigma v}{2} \hat{y} & \text{above bottom plate} \\ -\mu_0 \frac{\sigma v}{2} \hat{y} & \text{below bottom plate} \end{cases}$

$\Rightarrow \vec{B}_{\text{total}} = \begin{cases} (\mu_0 \sigma v)\hat{y} & \text{between the plates} \\ 0 & \text{otherwise} \end{cases}$

(b) Upper plate $\vec{K} = \sigma v \hat{x}$

Mag. field due to lower plate $\vec{B} = \mu_0 \frac{\sigma v}{2} \hat{y}$

$\Rightarrow$ force per unit area $\vec{f} = \vec{K} \times \vec{B} = \mu_0 \frac{\sigma^2 v^2}{2} \hat{z}$

(c) Electric field due to lower plate $\vec{E} = \frac{\sigma}{2\varepsilon_0}(-\hat{z})$.

$\Rightarrow$ force per area on upper plate $\vec{f}_{el} = \frac{\sigma^2}{2\varepsilon_0}(-\hat{z})$.

$0 = \vec{f} + \vec{f}_{el} \Rightarrow \frac{\mu_0 \sigma^2 v^2}{2} = \frac{\sigma^2}{2\varepsilon_0} \Rightarrow v^2 = \frac{1}{\varepsilon_0 \mu_0} = c^2$ ($c$ = speed of light)